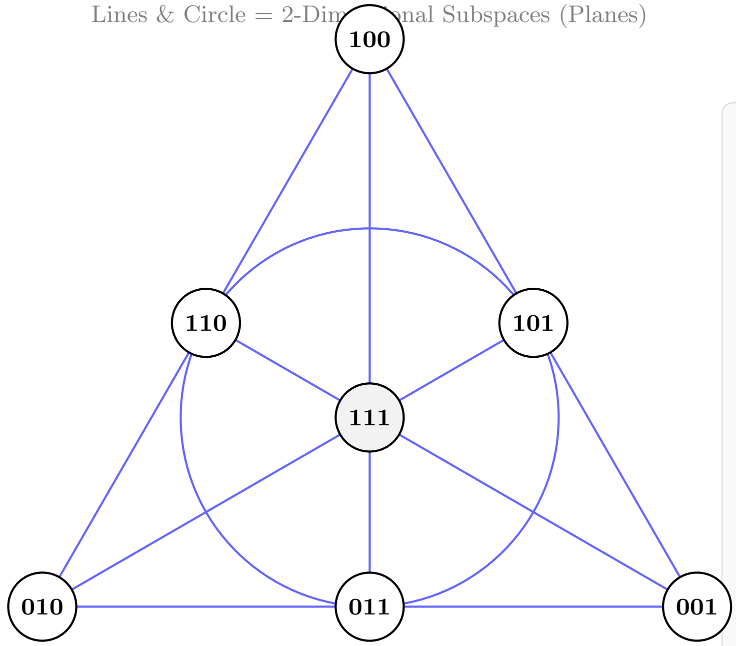
The Fano Plane: 7 Subspaces of $\mathbb{F}_2^3$

Points = 1-Dimensional Subspaces (Nonzero Vectors)
Lines & Circle = 2-Dimensional Subspaces (Planes)



Counting Ordered Bases vs. Subspaces

Numerator: Ordered Independent Pair

- 1st vector: 7 choices (any non-zero)
- 2nd vector: 6 choices (not in span of 1st)

$\Rightarrow 7 \times 6 = \mathbf{42}$ ordered pairs

Denominator: Bases per 2D Subspace

- A plane has 4 vectors (3 non-zero)
- 1st basis vector: 3 choices
- 2nd basis vector: 2 choices

$\Rightarrow 3 \times 2 = \mathbf{6}$ ordered bases per plane

Result: Total Distinct 2D Subspaces

$\begin{bmatrix} 3 \\ 2 \end{bmatrix}_2 = \dfrac{42}{6} = \mathbf{7}$ distinct planes